Final Project Presentation on

Solving TAPF via Poisson Potential Fields and Heuristic Word Reduction in Permutation Groups

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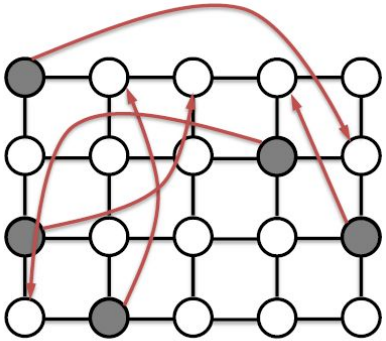
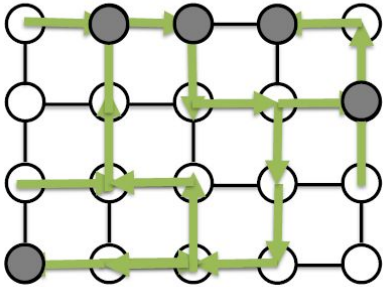
Overview

- **MAPF**: Multi-agent path finding from pre-assigned starts to goals
- **TAPF**: Assigning the starts and goals + MAPF
- **Problem**: MAPF/TAPF as graph traversal problem
- **Alternative**: MAPF/TAPF as **constrained word problem**
- **How**: Recasting problem as **Group Actions** of **Automorphism Groups**

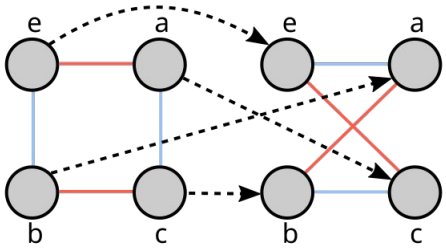
  Generated by Graph Edge Transpositions





Group Theory

- **Preliminaries**: Graph $G = (V, E)$, Agents A = {1, … , n}, Starts S, Goals G
- **Group** (H): Set with a binary operation that satisfies associativity, identity & inverse
- **Group Action** ($H \curvearrowright V$): A function that maps a group H and a set V to V itself
- **Automorphism Group** (Aut(V)): All bijective functions (permutations) from V to V
- **Generated by Graph Edge Transpositions** ($H = \langle (u\ v) \mid (u, v) \in E \rangle$): Composition of graph edges (adjacent transpositions) produce all members of group H
- **Constrained Word Problem**: Finding a constrained sequence of generator elements in H that compose together to generate the S to G permutation



$\phi(e) = e$
$\phi(a) = c$
$\phi(b) = a$
$\phi(c) = b$
$\phi = (acb)$

Group Theory (Formally)

A *group* $(G, *)$ is a set G together with an operation

$$*: G \times G \to G$$

such that

(a) $g * (h * k) = (g * h) * k$ for all $g, h, k \in G$;

(b) there exists an element $e \in G$ such that

$$g * e = e * g = g \qquad \text{for all} \quad g \in G;$$

(e is said to be an *identity element of* G);

(c) for every $g \in G$ there exists an element $g^{-1} \in G$ such that

$$g * g^{-1} = g^{-1} * g = e;$$

(g^{-1} is called an *inverse of* g).

Let G be a group, X a set. A G-*action on* X is a function

$$\begin{aligned} G \times X &\longrightarrow X \\ (g, x) &\longmapsto g \cdot x \end{aligned}$$

such that

(a) $e \cdot x = x$ for all $x \in X$;

(b) $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and all $x \in X$.

In this situation, we say that G *acts on* X and write $G \circlearrowright X$.

Related Works

- MAPF algorithms such as Joint-State A*, CBS, PBS, PIBT, LaCAM, etc
- TAPF algorithms such as CBM, CBS-TA, ITA-CBS, etc
- Group partitioning into Young subgroups via min-cuts (James, 1984)
- Symmetry groups and automorphism groups (Ghitza, 2021)
- Group actions (Rotman, 1965)
- Word problem solving via pebble motion (Kornhauser et al., 2015)
- Planning via navigation functions (Rimon & Koditschek, 1988)
- Poisson potential fields (Poisson, 1823)

Methodology (Local Heuristic)

- **Idea**: Greedy (fast) solver for constrained pebble moving problem
- Pick a single agent (pebble) to move
- Rank agent moves based on local heuristic (potential field)
- Move agent in preferred cell in the downward direction of potential field
- If the preferred cell is occupied, recursively displace the blocking chain to free the move
- If displacement is impossible, move to less preferred cells
- Restrict moves to non-backtracking moves to prevent oscillations

Pseudocode (Local Heuristic)

Algorithm 1 Low-Level Pebble Motion Step

```
L: current agent locations,  T: transposition sequence
Get legal moves 𝒩 of agent i
Rank 𝒩 and let best move be v
if v is free then
    Move i to v, update L, append (i, v) to T
else
    Let j be the agent blocking v
    if blocking agents can be recursively moved to free v then
        Recursively move blocking agents, update L, append transpositions to T
        Move i to v, update L, append (i, v) to T
    else if i has a valid non-backtracking move u then
        Move i to u, update L, append (i, u) to T
    else
        return failure
    end if
end if
```

Methodology (Global Guidance)

- **Idea**: No initial agent-goal allocation – let physics do the allocation for us
- Model agent locations as positively charged source particles
- Model goal locations as negatively charged sink particles
- Model map boundaries as Neumann boundary conditions (i.e. normal derivative is 0)
- Solve for Poisson's equation in free-space ($\nabla^2\varphi(x) = \rho$)
- Field lines of Poisson potential field will flow towards goals
- High-level search tree for agent-goal allocation
- Re-solve Poisson's equation whenever a goal is reached
- When agent-goal allocation is unsolvable, backtrack via UCB in search tree

Pseudocode (Global Guidance)

Algorithm 2 High-Level TAPF Solver

```
Initialize locations L ← S, moves T ← ∅, active agents A ← {1, ..., n}
Precompute potential field φ from active starts to active goals
while L ≠ G and time not exceeded do
    Choose agent i from A
    if L[i] ∈ G then
        Mark i inactive, update active goals, recompute φ
        Commit current moves T to the search tree
        continue
    end if
    if PebbleMotion(i, L, T, φ) succeeds then
        Update L and T
    else
        if backtrack by UCB succeeds then
            Restore previous state
            continue
        else
            break
        end if
    end if
end while
return Convert T to agent paths
```

Methodology & Pseudocode (Path Conversion)

- **Idea**: Convert pebble motion (group generator elements) into agent paths

Algorithm 3 Convert Word Sequence to Agent Paths

Require: Transposition deque T, number of agents n, start locations S
Ensure: Paths $\Pi = \{\pi_1, \ldots, \pi_n\}$

```
Initialize timestep deque 𝒯 ← ∅, current timestep τ ← ∅, used-agent set A ← ∅, used-
destination set D ← ∅
while T ≠ ∅ do
    Pop (i, (v_end, v_start)) from T
    if i ∈ A or v_end ∈ D then
        Push τ into 𝒯 and reset τ, A, D
    end if
    Add (i, v_end, v_start) to τ
    A ← A ∪ {i},    D ← D ∪ {v_end}
end while
Initialize each path π_i ← [s_i]
for each timestep τ in chronological order do
    Let M be the set of agents moving in τ
    for each (i, v_end, v_start) ∈ τ do
        Append v_end to π_i
    end for
    for each agent i ∉ M do
        Append the previous location of π_i to π_i
    end for
end for
return Π
```

Methodology & Pseudocode (Poisson Solver)

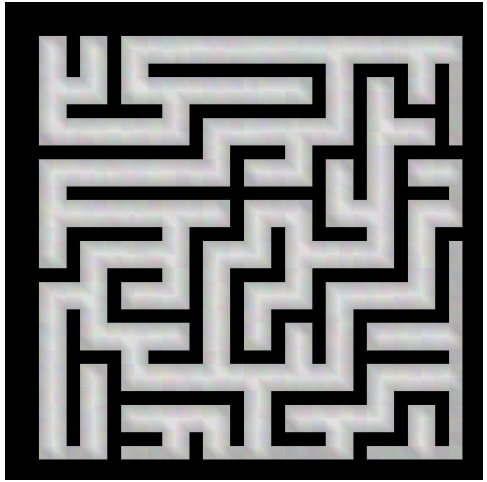
- **Idea**: Fast Poisson's equation solver in free-space using a sparse linear solver

Algorithm 4 Poisson Solver

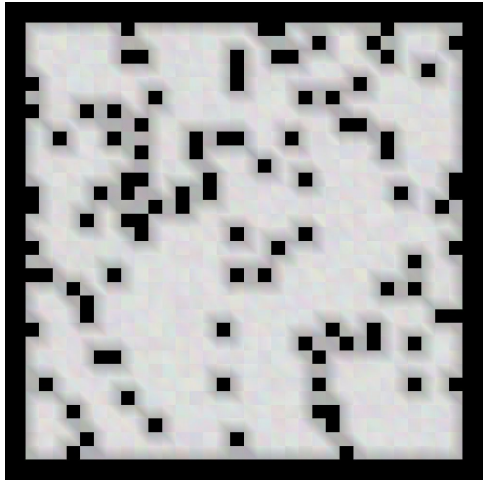
```
Pad map with obstacle boundary
Remove anchor-free connected free-space components
Flatten grid to indices k = r · w + c
for each free 4-neighbor pair (p, q) do
    Set A_pq ← −1
end for
for each cell p do
    if p is free and not fixed then
        Set A_pp ← deg(p)
    else
        Set A_pp ← 1
    end if
end for
Set b_p ← +α for starts, b_p ← −α for goals
Solve Aφ = b
Reshape, zero obstacle cells, and crop boundary
return φ
```

Baselines

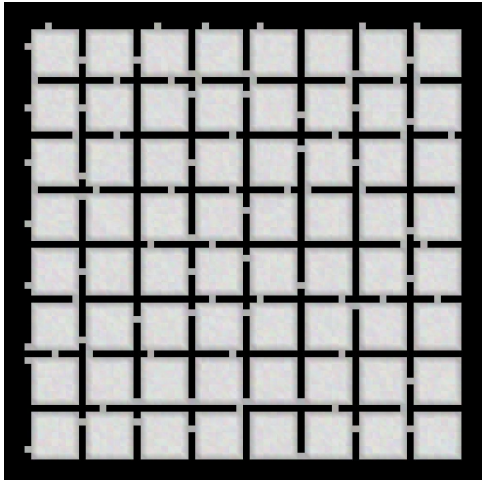
- Three maps from Moving AI Lab
- Agents and goals generated randomly in free-space
- Number of agents increased from 10 to 200 in 10-agent increments



maze-32-32-2



random-32-32-10

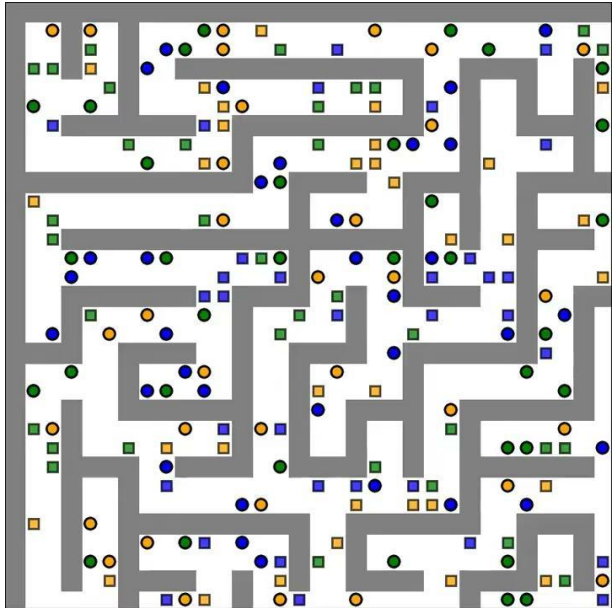


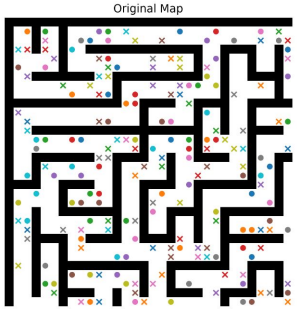
room-64-64-8

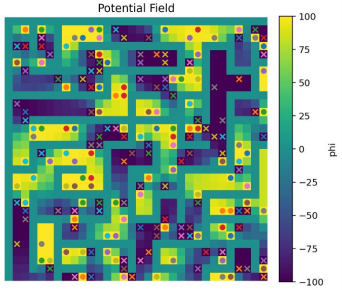
Results (maze-32-32-2)

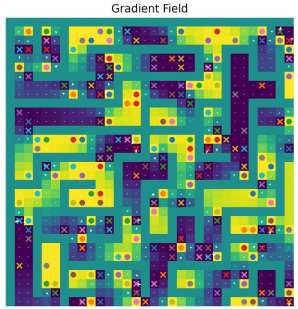
Table 1: Results on `maze-32-32-2`

# Agents	Cost	CPU Time (s)	Node Count
10	1190	0.018	13
20	3360	0.056	37
30	5310	0.089	59
40	5760	0.163	105
50	7700	0.196	107
60	10260	0.173	110
70	11130	0.219	142
80	10800	0.326	204
90	15030	0.351	219
100	19600	0.439	248
110	17050	0.392	247
120	28080	0.578	364
130	26520	0.603	359
140	38920	0.846	510
150	41250	0.893	526
160	33120	0.719	441
170	70550	1.426	898
180	63720	1.315	822
190	43700	0.961	555
200	37000	0.800	465





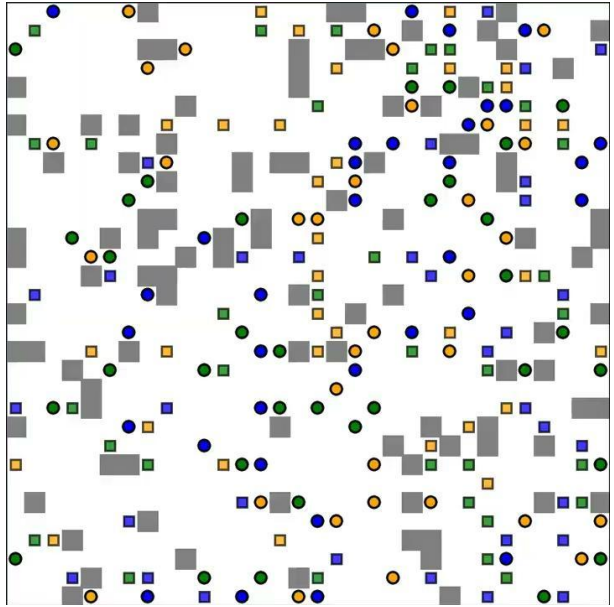


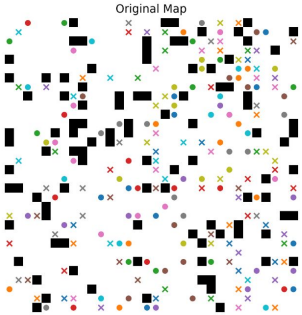


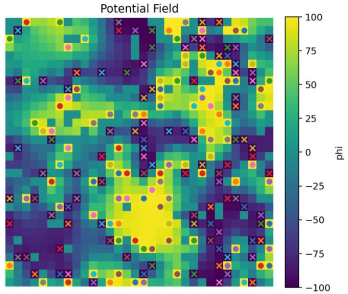
Results (random-32-32-10)

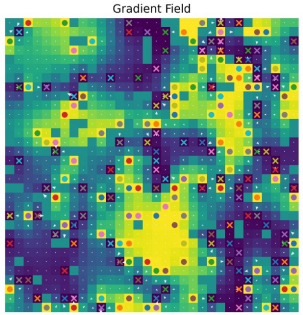
Table 2: Results on `random-32-32-10`

# Agents	Cost	CPU Time (s)	Node Count
10	420	0.024	11
20	1140	0.059	26
30	1560	0.086	37
40	3120	0.136	57
50	1700	0.154	64
60	3060	0.181	74
70	2800	0.213	87
80	4240	0.271	111
90	7560	0.387	145
100	8500	0.402	145
110	7260	0.392	152
120	9600	0.518	206
130	8580	0.545	217
140	11900	0.596	236
150	12300	0.666	262
160	12000	0.651	258
170	12750	0.722	282
180	18000	0.834	317
190	14250	0.801	305
200	16800	0.879	337





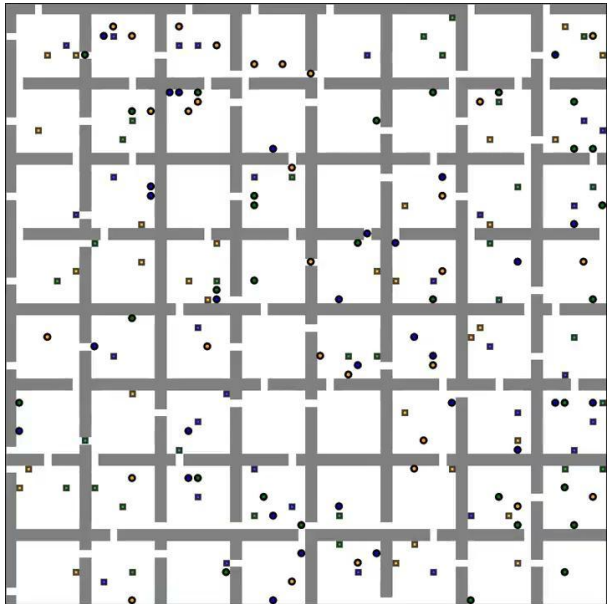


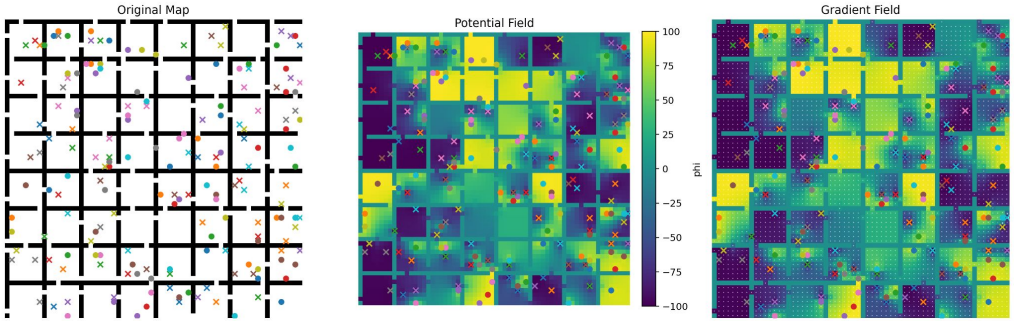


Results (room-64-64-8)

Table 3: Results on room-64-64-8

# Agents	Cost	CPU Time (s)	Node Count
10	1140	0.064	11
20	4280	0.241	32
30	5130	0.332	43
40	10920	0.923	116
50	8900	0.532	69
60	15360	1.541	200
70	19530	0.932	116
80	22320	0.975	121
90	19260	0.980	122
100	-1	30.003	4483
110	41250	1.559	186
120	25080	1.445	173
130	36140	1.976	234
140	57260	2.732	330
150	35250	1.989	236
160	44480	2.347	281
170	-1	30.002	4100
180	32220	2.133	256
190	42560	2.294	267
200	71800	3.237	391





Evaluation

- **Complete**: No
- **Optimal**: No
- **Speed**: Very fast
- Executor can automatically account for agent delays
  - If an agent is delayed, simply rerun path converter
  - Regroups pebble motion permutations into new blocks of agent paths
- Multiple agents can sometimes be attracted to the same sink. When the first agent occupies it, the recalculated potential field can cause the agents to travel great distances

Future Work

- Explore the method's theoretical properties
  - Intuition: Variations of this method could be probabilistic complete
  - Why: UCB incurs sub-linear regret – this class of algorithms tend to be probabilistic complete
  - Why: Physics-motivated – Maxwell's equations dictate that field lines eventually connect sources to sinks
- Multi-team TAPF with different agent and goal types
  - Intuition: Different potential fields for different group of agents
- Further group-theoretic MAPF/TAPF solvers

Conclusion

- Recast MAPF/TAPF problems as group action constrained word problem
- Solver combine global Poisson-field guidance with local greedy pebble motion
- High-level search tree with UCB backtracking for agent-goal allocation
- Fast runtime across multiple Moving AI benchmark maps
- No completeness or optimality guarantees
- Physics-inspired guidance and group-action structure can be a promising alternative to standard graph-search methods
- Future work on strengthening theory, improving allocation stability, and extend to multi-team TAPF

Thank You!

https://github.com/williamfbx/Group-Action-MAPF